# 1. Objective

Develop a simple web application to process, store, and search SWIFT MX messages. The application should allow users to log in, submit an MX XML message, view the extracted information, and search previously processed messages.

Candidates may select any suitable SWIFT MX message type for their implementation.

# 2. Functional Requirements

## A. Login

- Provide a login page with username and password.
- Authenticate users through the backend using predefined credentials.
- Display an appropriate error message for invalid credentials.
- Provide logout functionality.
- Protect message-related APIs from unauthenticated access.

## B. SWIFT MX Message Processing

- Allow users to enter or upload a sample SWIFT MX XML message.
- Parse the XML and extract relevant fields, such as message ID, sender, receiver, creation date, amount, currency, or transaction reference, wherever applicable.
- Display the extracted information.
- Save the original XML message and extracted fields in a database.
- Handle invalid XML and missing fields with meaningful error messages.

## C. Message Inquiry

- Display previously processed messages in a table.
- Allow users to search by message ID or other suitable fields.
- Display basic message information in the search results.
- Allow users to select a message and view its extracted fields and original XML.

# 3. Technical Requirements

- Frontend: ReactJS
- Backend: Java and Spring Boot

- Database: MySQL or PostgreSQL
- Communication: REST APIs
- XML parsing: Use a suitable XML parser and handle namespaces correctly.

# 4. Deliverables

- Frontend and backend source code.
- Database creation scripts.
- README with setup and execution instructions.
- One or more sample MX XML messages used for testing.

Candidates may use suitable libraries for XML parsing. The focus is on demonstrating XML processing and full-stack development skills.